Abstract Algebra II - Homework 3

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(a)

Let $f(x) = c_m x^m + c_{m-1}x^{m-1} + \cdots + c_1 x + c_0, c_i \in K \forall i \in \{0, 1, \ldots, m\}$. Let $\sigma \in G$ and $a_i \in X$, we have: (Clearly all elements in G fix K and $f(a_i) = 0$ and $\sigma(0) = 0$)

$$\begin{aligned}\sigma(f(a_i)) &= \sigma(c_m a_i^m) + \sigma(c_{m-1}a_i^{m-1}) + \cdots + \sigma(c_1 a_i) + \sigma(c_0) \\ &= c_m(\sigma(a_i))^m + c_{m-1}(\sigma(a_i))^{m-1} + \cdots + c_0 \\ &= f(\sigma(a_i)) = 0\end{aligned}$$

Thus, $\sigma(a_i)$ is also a root of f, so $\sigma(a_i) \in X$, since σ is a bijection on the field L, its restriction to the finite set X is a bijection from X to itself. Hence, every $\sigma \in G$ defines a permutation of X.

Define $\phi : G \to \text{Perm(X)},\ \sigma \in G$ by $\phi(\sigma) = \sigma|_X$ (the restriction of σ to the set X). For any $\sigma, \tau \in G$ and any $a \in X$, we have:

$$\phi(\sigma \circ \tau)(a) = (\sigma \circ \tau)(a) = \sigma(\tau(a)) = \phi(\sigma)(\phi(\tau)(a))$$

It's obviously a homomorphism. We then try to prove the map is injective, i.e., the kernel of ϕ is the identity automorphism. By definition, the splitting field L is the smallest field containing K and all the roots of f, therefore, L is generated by the roots over K, in other words, $L = K(a_1, a_2, \ldots, a_n)$.

Suppose $g \in \ker(\phi)$, any field automorphism in G is completely determined by its action on the generators of the field extension. Since g fixes K and fixes every generator a_i, it must fix every element of L, thus, g is the identity map. Because $\ker \phi = id_L$, the map ϕ is injective.

(b)

Theorem 1.1 (Isomorphism extension theorem)**.** *Given any field F, an algebraic extension field E of F and an isomorphism φ mapping F onto a field F' then φ can be extended to an isomorphism τ mapping E onto an algebraic extension E' of F' (a subfield of the algebraic closure of F').*

Theorem 1.2. *Let F be a field of characteristic 0 and let $f(x) \in F[x]$, and let G be the Galois group of $f(x)$ over F. If $f(x)$ is irreducible, G acts transitively on the roots of $f(x)$.*

Proof. Because f is an irreducible polynomial over K, for any two roots $a_i, a_j \in X$, there exists a field isomorphism between the simple extensions:

$$\theta : K(a_i) \to K(a_j)$$

that fixes K and maps a_i to a_j. Because L is the splitting field of f over both $K(a_i)$ and $K(a_j)$, this isomorphism θ can be extended to an automorphism σ of the entire splitting field L (by the Isomorphism Extension Theorem).

Therefore, there exists some $\sigma \in G$ such that $\sigma(a_i) = a_j$. This means the action of G on the set of roots X is transitive (there is only one orbit). □

By the orbit-stabilizer theorem, G acts on set X, the size of the orbit of an element multiplied by the size of its stabilizer subgroup equals the size of the group:

$$|G| = |\text{Orbit}(a_i)||\text{Stab}_G(a_i)|$$

Since the action is transitive, $|\text{Orbit}(a_i)| = |X| = n$, hence n divides $|G|$.

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Theorem 2.3. *Symmetric group S_p (where p is a prime) can be generated by a transposition and a p-cycle.*

Proof. Consider any transposition $\tau = (ab)$, and the p-cycle $\sigma = (12\dots p)$. Because $\langle\sigma\rangle$ is the subgroup of S_p consists of p-cycles, and every element has order p (p is prime) and can be the generator. Hence, there exists integer k such that σ^k sends a to b and is a p-cycle. □

To makes the notation easier, we can relabel the elements such that the position of a is marked as 1, and the position of $\sigma^k(a) = b$ is marked as 2 and so on. So the problem becomes finding the group generated by $(12, d \dots p)$ and (12). Since they can generate all kinds of adjacent transpositions, all transposition can be made by using adjacent transposition, all permutation can be made by muliple transpositions. Thus, (12) and $(12\dots p)$ generate S_p.

Let the Galois group of f be G.

By taking the derivative, $f'(x) = 5x^4 - 5 = 5(x-1)(x+1)(x^2+1)$, the local minimum of f is $f(1) = -5$ and the local maximum is $f(-1) = 3$. By some calculation we found: $f(0) = -1$, $\lim_{x\to\infty} f(x) = \infty$, $\lim_{x\to-\infty} f(x) = -\infty$. So $f(\infty)f(1) < 0$, $f(0)f(-1) < 0$, $f(-1)f(-\infty) < 0$, by intermediate value theorem, there exists at least one real roots on these intervals respectively: $(-\infty, -1)$, $(-1, 0)$, $(1, \infty)$, because the problem states that the real roots are less than 5, and the complex roots must be in conjugation forms (i.e. in pairs), so there are 3 real roots and 2 complex roots.

Let the complex roots be z and $\bar{z}$. The automorphism corresponding to complex conjugation fixes the 3 real roots and swaps z and $\bar{z}$. This is a transposition in G.

Since the G acts transitively on the roots of f, and since G is Galois, G is isomorphic to a subgroup of S_5. To maintain the group structure, we need to find transitive subgroup of S_5.

Consider a more general case, the galois group permutes p roots with p being prime. By orbit-stabilizer theorem, $|G|$ is divided by p. By cauchy's theorem, it contains a element of order p, which corresponds to the p-cycle in S_p. Here, we found G contains a 5-cycle.

By Theorem 2.3, since G contains both 5-cycle and a transposition, the only corresponding subgroup of S_5 is itself. As a result, the Galois group of f is isomorphic to S_5.